

MUTHUKKUMARAN B

Software Developer | Backend Developer | Web App Developer
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PROFESSIONAL SUMMARY

First-year Information Technology student with hands-on experience in Java, Python, C++, Data Structures, and Algorithms. Proficient in web app development. Strong problem-solving skills demonstrated through 50+ solved problems on LeetCode and HackerRank. Seeking a Software Developer Internship or Frontend Developer Internship to apply algorithmic knowledge, contribute to scalable software engineering projects, and grow within a high-performing engineering team.

TECHNICAL SKILLS

Programming Languages: Java, C++, C, Python, Django

Concepts: Object-Oriented Programming (OOP), Time and Space Complexity Analysis, CRUD Operations, REST APIs (Basic), Unit Testing

Tools & Platforms: Git, GitHub, VS Code, g++

PROJECTS

Air Canva - Virtual Drawing Application

December 2025

Python | Mediapipe | OpenCV | Numpy

- Developed a gesture-controlled virtual drawing application that allows users to draw on a digital canvas using hand movements captured through a webcam.
- Implemented real-time hand tracking and finger detection using MediaPipe to recognize gestures for drawing, erasing, and selecting colours.
- Utilized OpenCV for video capture and frame processing, enabling smooth real-time interaction and dynamic canvas rendering.
- Integrated NumPy for efficient image and pixel manipulation, improving drawing performance and responsiveness.

Platter to Purpose

February 2026

HTML5 | CSS3 | JavaScript | Python | Django | Git | GitHub

- Developed a full-stack community web platform that connects users with social resources and support opportunities.
- Built the backend using Python and Django, implementing dynamic routing, database interaction, and server-side logic.
- Designed a responsive and accessible front-end using HTML5, CSS3, and JavaScript following mobile-first and WCAG accessibility guidelines.
- Managed the complete development lifecycle using Git and GitHub, including version control, branching strategies, and collaborative commits.

Personal Portfolio Website

March 2026

HTML5 | CSS3 | JavaScript | Canvas API | Responsive Web Design

- Developed a responsive single-page portfolio website to present technical projects, skills, and achievements as an online professional profile.
- Designed modular sections including hero introduction, about, skills showcase, featured projects, resume download, and contact form with smooth scrolling navigation.
- Implemented interactive UI effects using JavaScript and the Canvas API, including an animated particle background with mouse-based interaction and hover animations.
- Built a mobile-first layout using semantic HTML5 and modern CSS3 techniques, integrating UI resources such as Font Awesome and Google Fonts to enhance visual design and usability.

EDUCATION

Bachelor of Technology - Information Technology

2025 - 2029

Sri Krishna College of Engineering and Technology, India | SGPA: 9.11 / 10.0

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming with C++, Discrete Mathematics, Computer Organization and Architecture

ACHIEVEMENTS AND COMPETITIVE PROGRAMMING

- Solved 50+ Data Structures and Algorithms problems on LeetCode and HackerRank, covering arrays, linked lists, trees, graphs, dynamic programming, and sorting.
- Maintained a 30-day consecutive problem-solving streak on LeetCode, logging daily solutions with time and space complexity annotations on GitHub.

CERTIFICATIONS

Advanced Diploma in Java Programming - Computer Software College

2025

INTERESTS

Competitive Programming | Data Structures and Algorithms | Backend Systems Design | Open-Source Software Development | Low-Level Programming | Distributed Systems | Software Engineering Research | Front End Development